Aaryan S. Maralihalli

Software & Machine Learning Engineer

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Summary

IEEE-published CS engineer who ships end-to-end — applied ML, backend, and customer-facing solutions. At **HyperVerge**, took 3 fintech KYC clients live at **>95% verification conversion (~10% lift)** across 15+ lines-of-business. Fine-tuned **Mistral-7B (LoRA)**, built a **secure RAG** service, and published interpretable cardiac ML (**ROC-AUC 0.757**) at IEEE ICASI 2026.

Experience

HyperVerge, Bangalore
Integration / Solutions Engineer Intern Jan 2026 – Jun 2026
AI identity-verification / KYC platform for banks & fintechs

- Owned end-to-end solutioning and integration for **6 enterprise fintech/lending clients** across **15+ lines-of-business** — CKYC, Video KYC/VPD, eSign, Name Match, Liveness, Face Match, DigiLocker, OCR/NSDL-PAN.
- Took **3 clients live at >95% verification conversion** (~10% lift) by matching pain points to the right modules and proactively adding ones they hadn't scoped.
- Authored Solution Design Docs, ran client integration calls and go-lives, issued production credentials, and drove post-go-live **BI & stabilization**.
- Built **"HV Deals Bot"** (Python + Slack) to track deal stages and document the delivery process for new joiners.

Integrated Genetic Solutions (IGS), Bangalore
Project Intern, Medical AI & Web Dev Jun – Aug 2024

- Migrated a legacy Bootstrap-JS dashboard to **React (MERN)**, improving processing of **5M+ blood-lab values** on a HIPAA-compliant medical-AI product.

Selected Projects

Self-Supervised Cardiac Model (IEEE-published) github.com/Aaryansm11/ecg-ppg-arrhythmia-prediction

- Contrastive learning (SimCLR, NT-Xent) on MIMIC-III ECG/PPG; 1D-CNN encoder + explainability; ROC-AUC 0.757, benchmarked vs CHA2DS2-VASc.

LANGTRON — AI technical-memory assistant github.com/Aaryansm11/LANGTRON

- Fine-tuned Mistral-7B (LoRA) on 30K technical Q&A; dual-model architecture (retrieval + problem-solving); FAISS semantic memory with sub-100ms retrieval; 100% local.

SDIS — Secure Document Intelligence Service github.com/Aaryansm11/SDIS

- FastAPI RAG microservice — JWT/RBAC, PII redaction (spaCy), immutable audit logging, FAISS over PostgreSQL; Dockerized on AWS ECS; pytest + GitHub Actions CI.

our-life-hq — 3D full-stack app github.com/Aaryansm11/our-life-hq

- Interactive React Three Fiber front-end with Supabase auth/data, deployed on Cloudflare.

Live Emostream — real-time pipeline github.com/Aaryansm11/Live_emostream

- Event-driven Kafka + Spark + Redis with nginx load balancing; load-tested to 1,000 concurrent users (Locust).

Skills

- **Languages:** Python, JavaScript (React), SQL, Java
- **ML / AI:** PyTorch, Hugging Face Transformers, LLM fine-tuning (LoRA), RAG, LangChain, FAISS, spaCy, BERT, scikit-learn, contrastive / self-supervised learning, explainable AI
- **Backend / Data:** FastAPI, Flask, Django, REST, JWT, PostgreSQL, MySQL, Neo4j, Redis, Kafka, Spark, Docker, AWS (EC2/S3/IAM/ECS), GitHub Actions (CI)
- **Solutions & CS:** solution design, client integration & go-live, KYC / identity domain, BI & stabilization; DSA (150+ LeetCode), OOAD, DBMS

Publication & Education

- **IEEE ICASI 2026 (Kyoto), Oral Presentation** — "An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG."
- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026
- **Certifications:** Graphics Programming with WebGL (PESU I/O); Problem Solving — Intermediate (HackerRank); CITI Program